

PRACTICAL 4

Practical4(1)

Problem Phase

A hospital manages a queue of patient tokens. Critical patients are added at the front, routine patients are added at the end, and priority patients can be inserted at a specific position.

Solution

- Use `push_front()` to add a critical patient.
- Use `push_back()` to add a routine patient.
- Use an iterator and `insert()` to add a patient at a specific position.
- If the position is greater than the queue length, add the patient at the end.

Practical4(2)

Problem Phase

Remove a patient token when a patient leaves, display the queue from front to back, and print the remaining tokens from last to first.

Solution

- Use `remove()` to delete a patient by value.
- Use a normal iterator for forward traversal.
- Use a reverse iterator for reverse printing.